STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-PRODUCT-K-INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3967-3976 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3978-3992, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 44 of queue_sdga.json (pb=531, stmt 3967-3976, proof 3978-3992).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-product-K-injective' also exists in stacks-derived — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let T be a set and for each $t \in T$ let $\mathcal{I}_t$ be a K-injective diffential graded $\mathcal{A}$-module. Then $\prod \mathcal{I}_t$ is a K-injective differential graded $\mathcal{A}$-module.*

Proof. Let $\mathcal{K}$ be an acyclic differential graded $\mathcal{A}$-module. Then we have

$$\mathrm{Hom}_{Mod^{dg}(\mathcal{A},\mathrm{d})}(\mathcal{K}, \prod\nolimits_{t\in T} \mathcal{I}_t) = \prod\nolimits_{t\in T} \mathrm{Hom}_{Mod^{dg}(\mathcal{A},\mathrm{d})}(\mathcal{K}, \mathcal{I}_t)$$

because taking products in $Mod(\mathcal{A}, \mathrm{d})$ commutes with the forgetful functor to graded $\mathcal{A}$-modules. Since taking products is an exact functor on the category of abelian groups we conclude. □

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.